

SecureMailScope forensic report

cleartext_auth.pcap

SHA-256: a8416ab9dedca26247afe2c0bc54492afb9c5a897dd7132f7189f8525fbafb9d

Overall posture: 47.7/100

smtp-stream-0

192.0.2.10 → 198.51.100.25:587 | Score: 43

ML risk: Critical; anomaly: False. synthetic_training_demo; normalized features only.

STARTTLS-002 / Medium: Advertised STARTTLS not used

The observed session did not upgrade.

Frame 6, timestamp 1789812000.05058

Enforce STARTTLS on client and server.

AUTH-001 / Critical: Authentication exposed in plaintext

Authentication was attempted before encryption. Credential content is redacted.

Frame 7, timestamp 1789812000.06782

Require TLS before AUTH, LOGIN or PASS; rotate exposed credentials.

Score deductions: [{"source": "rule", "id": "STARTTLS-002", "severity": "Medium", "deduction": 7}, {"source": "rule", "id": "AUTH-001", "severity": "Critical", "deduction": 30}, {"source": "ml", "id": "Critical", "deduction": 20}]

Frame 4 [server] 220 [response content omitted]

Frame 5 [client] EHLO [REDACTED]

Frame 6 [server] 250 [response content omitted]

Frame 6 [server] 250 STARTTLS advertised

Frame 7 [client] AUTH [REDACTED]

Frame 8 [server] 235 [response content omitted]

Frame 9 [client] QUIT

imap-stream-1

192.0.2.10 → 198.51.100.25:143 | Score: 50

ML risk: Critical; anomaly: False. synthetic_training_demo; normalized features only.

AUTH-001 / Critical: Authentication exposed in plaintext

Authentication was attempted before encryption. Credential content is redacted.

Frame 16, timestamp 1789812000.137135

Require TLS before AUTH, LOGIN or PASS; rotate exposed credentials.

Score deductions: [{"source": "rule", "id": "AUTH-001", "severity": "Critical", "deduction": 30}, {"source": "ml", "id": "Critical", "deduction": 20}]

Frame 15 [server] * [response content omitted]

Frame 16 [client] LOGIN [REDACTED]

Frame 17 [server] * [response content omitted]

Frame 18 [client] LOGOUT

pop3-stream-2

192.0.2.10 → 198.51.100.25:110 | Score: 50

ML risk: Critical; anomaly: False. synthetic_training_demo; normalized features only.

AUTH-001 / Critical: Authentication exposed in plaintext

Authentication was attempted before encryption. Credential content is redacted.

Frame 27, timestamp 1789812000.285044

Require TLS before AUTH, LOGIN or PASS; rotate exposed credentials.

Score deductions: [{"source": "rule", "id": "AUTH-001", "severity": "Critical", "deduction": 30}, {"source": "ml", "id": "Critical", "deduction": 20}]

Frame 24 [server] +OK [response content omitted]

Frame 25 [client] USER [REDACTED]

Frame 26 [server] +OK [response content omitted]

Frame 27 [client] PASS [REDACTED]

Frame 28 [server] +OK [response content omitted]

Frame 29 [client] QUIT

Assessment limits

Scores describe observed evidence, not proof of endpoint security. Missing evidence is not a passing certificate check.

TLS 1.3 certificates and encrypted authentication are not visible without session secrets. No decryption is attempted.

Revocation is unknown offline. Chain validation uses bundled certifi roots and capture time; private roots may be untrusted.

ML is a synthetic-data demonstration, not a validated production detector. Only normalized cryptographic features are used.

Partial streams are excluded from overall posture. Conflicting TCP overlaps retain first observed bytes and are flagged.

Scapy extracts TCP; IP fragment reassembly, SSLv2 decoding, QUIC, and arbitrary application protocols are not supported.